Vijay Tudimilla

Python Developer

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Professional Summary:

Highly skilled **Python Developer** with 4+ years of experience in designing, developing, and deploying scalable backend systems, REST APIs, and data automation solutions across finance, healthcare, and analytics domains. Expert in building robust **Python**, **Flask**, and **FastAPI** applications, integrating **data engineering pipelines**, and implementing **cloud-native deployments** using **AWS** and **Azure**. Adept at writing optimized SQL, developing ETL workflows, and automating tasks using **Airflow**, **Kafka**, and **Docker**. Strong advocate of clean code, performance tuning, and collaboration in **Agile/Scrum** environments. Passionate about bridging backend development and data integration for enterprise-grade digital solutions.

Technical Skills:

Programming Languages: Python, SQL, JavaScript, R, Shell Scripting
Frameworks & Libraries: Flask, FastAPI, Django, RESTful APIs, Microservices
Data Engineering: Pandas, NumPy, Apache Airflow, PySpark, Kafka, ETL Design, Feature Engineering
Databases: MySQL, PostgreSQL, MongoDB, Oracle, Snowflake, Redis
Visualization & BI: Power BI, Tableau, Matplotlib, Seaborn, Plotly, Dash
Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), Azure, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD Pipelines.
Version Control: Git, GitHub, GitLab, Bitbucket
Operating Systems: Linux, Unix, Windows
Methodologies: Agile, Scrum, SDLC, TDD

Professional Experience:

Client: Fannie Mae –Chicago, IL.
Role: *Python Developer* June 2024 – Present
Responsibilities:

- Built Python-based APIs using FastAPI to support internal applications with real-time scoring, analytics, and data services; improved API responsiveness by ~25% through async processing and response optimization.
- Developed Python ETL workflows in Airflow to clean, validate, and transform mortgage datasets into structured feature layers used by reporting tools and ML models.
- Created Python Kafka consumers/producers to process streaming loan events and feed pipelines that support monitoring, scoring, and event-driven automation.
- Integrated Python microservices with registered ML models deployed via MLflow/SageMaker, ensuring safe model versioning, input checks, and seamless fallbacks.
- Added LLM-powered document extraction capabilities to backend services using Python (LangChain + HuggingFace/OpenAI) to support automated review of loan documents.
- Optimized Python + SQL data processing tasks by tuning queries, batching operations, and indexing tables, reducing pipeline runtimes and cloud compute usage by ~30%.
- Built Python utilities and internal command-line tools to streamline debugging, data checks, and one-off operational tasks, reducing manual workload for multiple teams.
- Implemented unit tests, logging standards, and error-handling patterns across Python services to improve reliability and maintainability.
- Collaborated with front-end teams, analysts, and data engineers to align API behavior, payload structures, and integration workflows across systems.

Tools & Tech: Python, Flask, FastAPI, SQL, Pandas, Airflow, Kafka, Power BI, AWS, Jenkins, GitHub

Client: ResMed – Bentonville, AR.

Role: *Python Backend Developer* Jan 2024 – May 2024

Responsibilities:

- Developed Python REST APIs with Flask to provide demand forecasts, recommendations, and analytics outputs to retail and e-commerce applications.
- Built Python-driven forecasting pipelines (LSTM + Prophet) orchestrated via Airflow, delivering SKU- and store-level predictions that reduced stock-outs by ~30%.
- Implemented a Python-based recommendation engine using PyTorch/Scikit-learn, improving product conversion rates by 18% during A/B testing.
- Performed Python-based feature engineering and dataset preparation using Snowflake SQL and Pandas to support accurate model training and scoring.
- Integrated AI-driven personalization flows using Python OpenAI clients to generate context-aware descriptions and recommendations for marketing workflows.
- Containerized Python services using Docker and deployed them to AWS EKS with CI/CD pipelines powered by GitHub Actions.
- Developed Python automation scripts to calculate model performance metrics and generate visualizations consumed by Tableau and Power BI dashboards.
- Optimized Python data loaders, inference logic, and SQL queries to improve API latency and ensure smooth integration with JavaScript front-end modules.
- Worked closely with product, analytics, and engineering teams to refine requirements, align contract schemas, and ensure robust Python service behavior in production.

Tools & Tech: Python, Flask, Docker, Kubernetes, SQL, Kafka, Airflow, AWS ECS, Tableau, Jenkins

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Client: TabCaps Softtech (Contract) – Hyderabad, India

Role: *Junior Python Developer* Sep 2021 – Aug 2023

Responsibilities:

- Developed automation scripts in Python to streamline Excel-based reporting and data cleaning tasks.
- Built Power BI dashboards for tracking operational metrics and performance KPIs.
- Assisted in backend microservice development using Flask and Django frameworks.
- Wrote complex SQL queries and functions for business data analysis.
- Conducted data validation and created data dictionaries for ongoing projects.
- Implemented logging and exception handling within ETL pipelines to improve maintainability.
- Collaborated with DevOps to automate deployment pipelines on AWS.
- Assisted in the integration of APIs for machine learning model consumption in production.
- Built scripts to extract and process JSON and CSV data files from multiple sources.
- Supported cross-functional teams in Agile sprints and delivered assigned features on time.
- Participated in peer code reviews and contributed to improving development standards.
- Authored clear documentation for all automation workflows and dashboards.

Tools & Tech: Python, Flask, Django, SQL, Power BI, AWS, Excel, Pandas

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Projects:

- **Fraud Detection API:** Built scalable REST APIs using Flask and FastAPI to deliver real-time fraud risk scores to internal systems and BI dashboards. Integrated Python scoring pipelines with ML models, added request validation and logging, and optimized response execution for high-volume transaction loads.

- **Data Automation Suite:** Developed automated ETL pipelines using Python, Airflow, and Kafka to orchestrate near real-time data ingestion, transformation, and quality checks. Enabled continuous data availability for analytics teams by implementing event-driven workflows and centralized data validation logic.
- **Inventory Optimization System:** Designed Python-based backend services using Flask to support retail demand forecasting and inventory optimization workflows. Integrated forecasting outputs with Power BI dashboards, enabling business teams to visualize SKU-level predictions and improve replenishment accuracy.

Education:

- Master's in computer technology from Eastern Illinois University, Charleston, Illinois, USA.
- Bachelor of Technology in Engineering from Jawaharlal Nehru Technological University, Hyderabad.

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